

Institute of Engineering & Management

Department of Electrical Engineering

Report on the Five-Day Faculty Development Programme (FDP)

on “**Synthesizing Classical and Modern Approaches in Robotics, Control Systems, and Industrial Automation**”

13th – 17th October 2025 (Online Mode)

1. Introduction

The Department of Electrical Engineering, Institute of Engineering & Management (IEM), successfully organized a **Five-Day Faculty Development Programme (FDP)** titled “*Synthesizing Classical and Modern Approaches in Robotics, Control Systems, and Industrial Automation*” from **13th to 17th October 2025** in **online mode**.

The FDP aimed to bring together academicians, researchers, and industry experts to discuss and explore the latest trends, methodologies, and technologies in robotics, control systems, and industrial automation. The program focused on integrating classical control theories with modern intelligent and data-driven approaches, offering participants a well-rounded perspective on current and future research directions.

2. Objectives of the FDP

The primary objectives of the programme were:

- To enhance the technical knowledge of faculty members in advanced control and automation systems.
 - To familiarize participants with cutting-edge developments in robotics and industrial applications.
 - To encourage collaborative learning and knowledge sharing among researchers and practitioners.
 - To bridge the gap between academic research and industry requirements through expert interactions.
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3. Programme Overview

The five-day FDP consisted of **expert sessions by eminent speakers** from reputed institutes and industries across India and abroad. Each session included interactive discussions and Q&A segments to engage participants effectively.

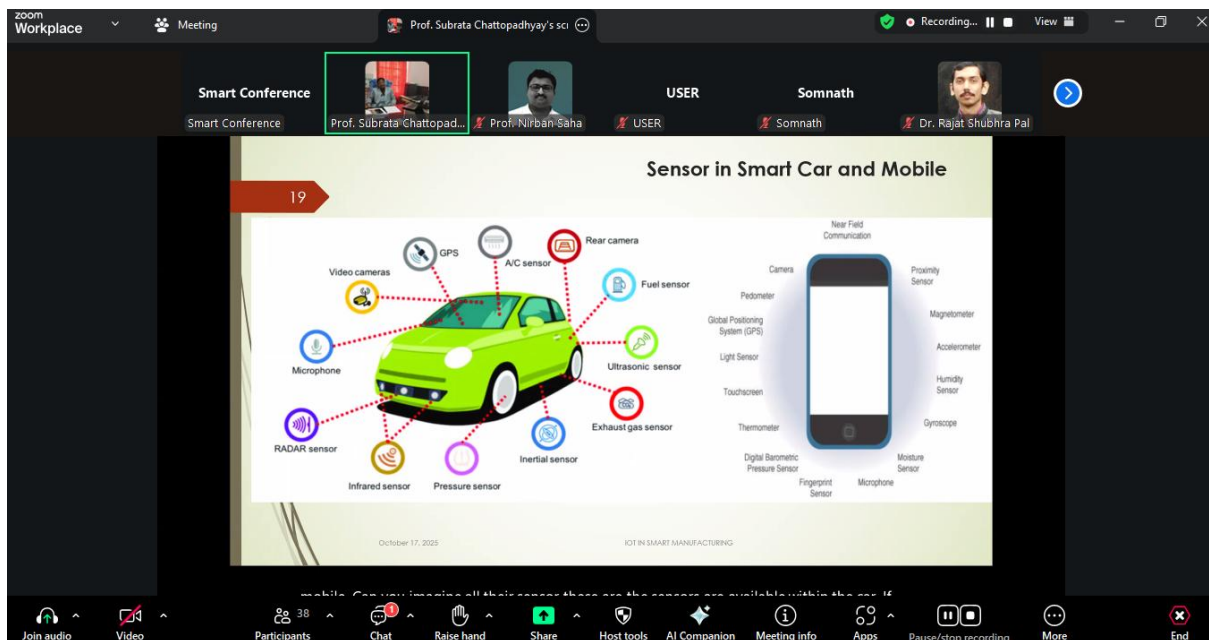
4. Day-wise Summary of Sessions

Day 1 – 13th October 2025

Speaker: Dr. Subrata Chattopadhyay, Professor, NITTTR Kolkata

Topic: IoT in Manufacturing Industry

The FDP commenced with an inaugural session, followed by a lecture by Dr. Chattopadhyay. He elaborated on the role of the Internet of Things (IoT) in enhancing manufacturing processes, emphasizing smart factories, predictive maintenance, and data-driven decision-making. Participants gained valuable insights into how IoT bridges automation with analytics.



Day 2 – 14th October 2025

Speaker: Dr. Falguni Gopmandal, Development Specialist, KPIT Technologies Ltd., Bangalore

Topic: Robust Multivariable PID Controller Design for Uncertain Systems with Application to Advanced Driving Assistance Systems

Dr. Gopmandal discussed modern PID control approaches for uncertain systems, with special reference to automotive applications. Her session provided a comprehensive understanding of robust control techniques used in Advanced Driving Assistance Systems (ADAS).

Uncertain Plant Description

- The uncertain plant

$$\dot{x} = A(t)x + B_w(t)w + B_u(t)u, \quad z = C_zx + D_{zw}w + D_{zu}u, \quad y = C_yx$$

$$A(t) = A_0 + \Delta A(t), \quad B_w(t) = B_{w0} + \Delta B_w(t), \quad B_u(t) = B_{u0} + \Delta B_u(t)$$

$$\begin{bmatrix} \Delta A(t) & \Delta B_w(t) & \Delta B_u(t) \end{bmatrix} = \begin{bmatrix} D_1 & D_2 & D_3 \end{bmatrix} \begin{bmatrix} \Delta_1(t) & 0 & 0 \\ 0 & \Delta_2(t) & 0 \\ 0 & 0 & \Delta_3(t) \end{bmatrix} \begin{bmatrix} E_1 & 0 & 0 \\ 0 & E_2 & 0 \\ 0 & 0 & E_3 \end{bmatrix} \quad (1)$$

- Time varying norm-bounded uncertain matrices satisfy

$$\|\Delta_1(t)\|_2 \leq 1, \quad \|\Delta_2(t)\|_2 \leq 1, \quad \|\Delta_3(t)\|_2 \leq 1 \quad (2)$$

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Response in Low Speed Maneuvering

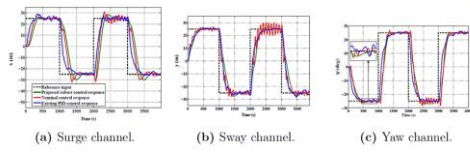
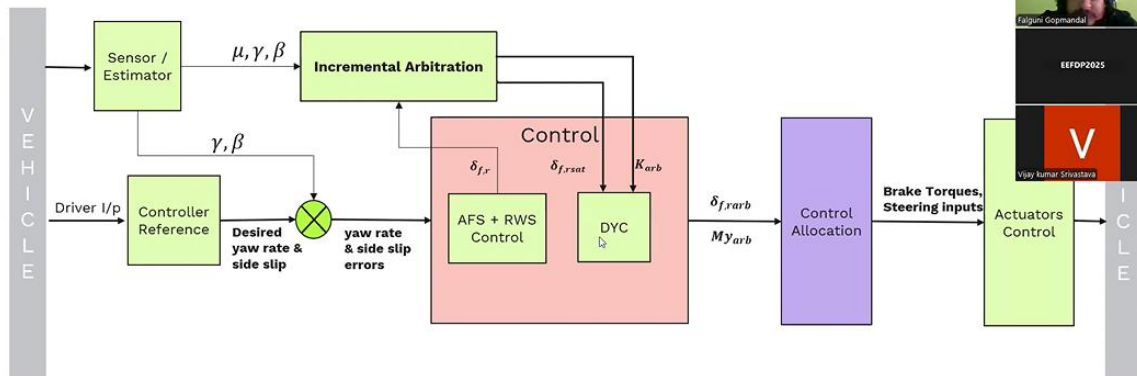


Figure 11: Closed-loop response of uncertain system in low speed maneuvering.

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Incremental Arbitration Logic Architecture



KPIT

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Day 3 – 15th October 2025

Speaker: Dr. Manas Kumar Bera, Associate Professor, NIT Rourkela

Topic: Event-Based Sliding Mode Control of Mobile Robots

Dr. Bera presented an insightful session on event-driven control strategies in mobile robotics. He demonstrated how sliding mode control can improve system robustness and performance while minimizing computational effort—an essential aspect of real-time robotics applications.

Manipulator Vs Mobile Robot

Manipulator	Mobile Robot
• Operate in a constrained workspace.	• Can operate in an unconstrained environment.
• Have absolute measurements of position.	• Need external sensing to determine position.
• May or may not need to perceive the world around them.	• Need external sensing to avoid obstacles.

• A mobile manipulation system offers a dual advantage of manoeuvrability offered by the mobile base and manipulability or dexterity offered by the manipulator. Further, the mobile base offers unlimited workspace to the manipulator.

AMRs are used to transport components from the warehouse to assembly lines

Participants (49): SC S... (Host, me, Participant ID: 495078), DM Dr. Manas Kumar Bera (Co-host), AS Abhisek Satapathi (Co-host), Prof. Nirban Saha (Co-host), 91801, Ami Barman, AG Anjana Goen, Ankita, AM Aparajita Mukherjee, ARITRA, Ashis Das, CK Chandra Kumar Adak, Nehaniana Datta Mitra.

Day 4 – 16th October 2025

Speaker: Dr. Debdoot Sain, Postdoctoral Researcher, Division of Electronic Engineering, Jeonbuk National University, South Korea

Topic: Interval Type-2 Fuzzy PID Controllers with Interval of Confidence and Various Footprints of Uncertainty: From Theory to Experimental Validation

Dr. Sain's talk bridged theoretical and experimental perspectives on fuzzy control systems. His detailed discussion on uncertainty modeling and experimental validation techniques provided deep academic value to the participants.

Canonical Nonholonomic Robot Model

Theorem
The system $\dot{q} = G(q)u$, where $\text{rank } G(0) < \dim(q)$, cannot be stabilized to the origin by a continuous time-invariant feedback control law.

Conclusion

- The nonholonomic robot is not linearly controllable. In fact, there is no controller that is a continuous function of q which can stabilize $\dot{q} = 0$.
- This theorem applies to our canonical nonholonomic robot model, since the rank of $G(q)$ is 2 everywhere (there are only two control vector fields), while the chassis configuration is three-dimensional.
- Even though the canonical nonholonomic robot is not linearly controllable, it still satisfies other important notions of controllability.
- In particular, the velocity constraint does not integrate to a configuration constraint-the set of reachable configurations is not reduced because of the velocity constraint.

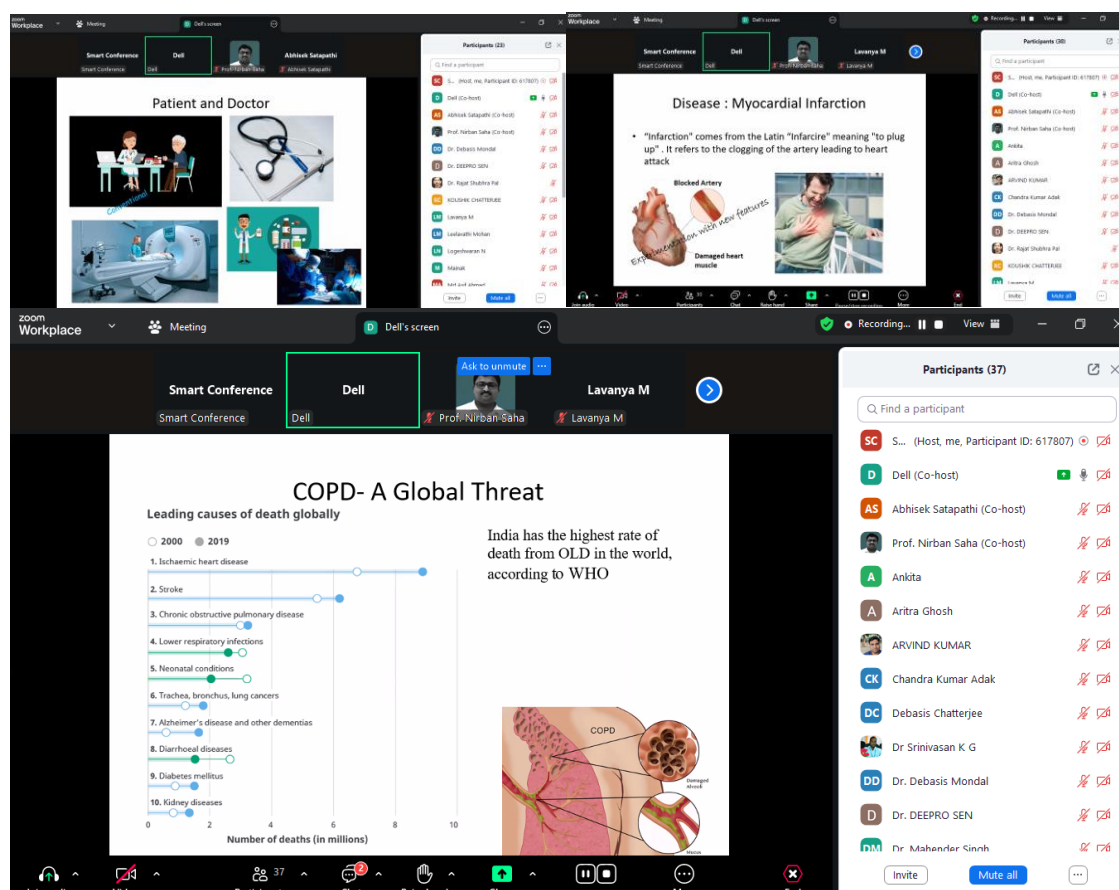
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Day 5 – 17th October 2025

Speaker: Dr. Saurabh Pal, Associate Professor, Department of Applied Physics, University of Calcutta

Topic: *Recent Trends in Biomedical Signal and Image Processing and Analysis*

The final day's session explored applications of automation and signal processing in biomedical engineering. Dr. Pal illustrated emerging techniques in data acquisition, filtering, and analysis of biomedical signals, highlighting interdisciplinary connections between electrical engineering and medical technology.



5. Participation and Engagement

The FDP witnessed enthusiastic participation from **faculty members, research scholars, and industry professionals** from across the country. Interactive Q&A sessions, online discussions, and feedback interactions created an engaging and productive learning environment. The sessions were conducted via the Zoom platform, ensuring accessibility and flexibility for participants.

6. Outcomes of the Programme

- Enhanced understanding of advanced control systems, robotics, and automation concepts.
- Exposure to real-world industrial applications and research methodologies.
- Strengthened collaboration between academia and industry experts.
- Motivated participants to explore interdisciplinary research in automation and intelligent systems.

All participants received **E-certificates** upon successful completion of the programme.

7. Acknowledgements

The Department of Electrical Engineering extends heartfelt gratitude to all distinguished speakers for their valuable time and contributions. Special thanks to the organizing committee members, technical team, and participants for making this FDP a grand success.

The department also acknowledges the support and encouragement provided by the management of the Institute of Engineering & Management, which played a vital role in the smooth execution of this event.

8. Conclusion

The **5-Day Faculty Development Programme (FDP)** on “*Synthesizing Classical and Modern Approaches in Robotics, Control Systems, and Industrial Automation*” concluded successfully on 17th October 2025. The programme enriched participants with a blend of theoretical insights and practical applications, aligning with the institute’s continuous endeavor to foster research and innovation among faculty members.